

Annexure-I

Hackathon Challenge — Project NETRA

Networked Eyes for Tactical Response and Awareness A Citizen-Consented, Camera-Agnostic, Hybrid Video-Intelligence Platform for Public Safety

(Netra — “eye” in Sanskrit and most Indian languages — emphasizes that the system extends the eyes of well-meaning citizens to the public-safety apparatus, never the reverse.)

0. Critical Framing — Read Before Building

This challenge stands on a knife-edge. A well-designed citizen-camera-sharing platform makes communities safer, recovers missing children, solves hit-and-runs, and helps small businesses get justice after burglaries. A poorly-designed one creates a permanent panopticon, chills public life, enables stalking and political surveillance, and dies in court. Recent Indian deployments illustrate both sides:

The Telangana State Police facial-recognition rollout and the Delhi Police FRS have faced sustained constitutional challenges from the Internet Freedom Foundation and others on grounds of absent legal basis, opacity, and demographic bias.

The Hyderabad “Nenu Saitham” / Community CCTV programmes have shown that citizen-shared cameras genuinely help solve crime when governance is tight and scope is bounded.

The difference between those outcomes is not the technology. It is the governance, the consent architecture, the bystander-protection layer, and the discipline of purpose limitation. Teams that internalize this will build systems that get deployed. Teams that don't, will build systems that get litigated out of existence.

The four design rules that follow are non-negotiable:

1. Citizen-pull, not police-push. Citizens onboard their own cameras. Citizens revoke instantly. The system never auto-discovers, scrapes, or accesses any camera without explicit per-camera consent.

2. Bystanders matter as much as participants. Most faces and vehicles in any citizen feed belong to people who never consented. The system must protect them as a first-class concern — privacy zones, FOV validation, default blurring of incidental subjects, and a strict separation between general indexing and targeted recognition.

3. Face recognition is a lead-generation tool, not an identification tool. No autonomous identifications. No single-match results. Mandatory top-N candidates with calibrated confidence. Mandatory human verification. Watchlists are governed, not curated by line officers.

4. Hybrid by default, live by exception. Event clips are the default uplink. Live pulls require authorization and are logged. Bandwidth, privacy, and proportionality all point the same way.

§3 of this document operationalizes these rules. They are gating conditions in the judging rubric.

1. Background & Context

India's CCTV ecosystem is bifurcated:

State-deployed CCTV in major cities is concentrated in Smart-City Integrated Command and Control Centres (ICCCs) — Hyderabad, Bengaluru, Pune, Surat, Indore, and ~100 others. These cover arterial roads, public squares, and government buildings, but rarely the dense interiors of residential colonies, markets, alleys, and intra-city by-lanes where most non-traffic crime occurs.

Citizen and commercial CCTV — installed by RWAs, shop-owners, small businesses,

individual homeowners, and increasingly by dashcam users — covers exactly those interior spaces, but is invisible to law enforcement. After every incident, the IO's first task is to physically walk the locality knocking on doors and shops to ask if anyone has footage, often arriving days after the footage has already been overwritten.

The technology gap is striking. A typical urban Indian neighbourhood has 50–200 citizen-owned cameras within a 500 m radius — IP cameras, DVR-based analog systems, Tapo and Mi consumer devices, dashcams parked overnight, and increasingly smartphone-mounted “doorbell” rigs. Their feeds are siloed on home Wi-Fi, in vendor clouds (Hikvision iVMS, Dahua DMSS, Tapo Care), or on DVRs with 7–14 day overwrite cycles. The data exists. The bridge to police use does not.

A citizen-consented, opt-in, camera-agnostic platform can close this gap — provided it is built with privacy, bystander protection, watchlist governance, and mission-creep prevention as load-bearing structural elements, not bolt-ons.

2. The Problem Statement

There is no production-grade, vendor-agnostic, citizen-consented platform that:

1. Onboards arbitrary heterogeneous citizen-owned cameras (IP/RTSP/ONVIF, vendor-proprietary cloud cameras, consumer Wi-Fi cameras, DVR-attached analog cameras, dashcams, and smartphone cameras) into a unified ingestion fabric;
 2. Operates in a bandwidth-rational hybrid mode — event-triggered clip uploads by default, authorized live pull-on-demand by exception, all geo-mapped and time-indexed;
 3. Provides police Control Room operators a map-first investigation console with retrieval by attribute, by ANPR plate, by face (against authorized watchlists only), by event type, and by cross-camera trajectory;
 4. Generates a defined set of operationally useful alerts (night-time motion in away-mode premises, vehicle of interest, person of interest from authorized watchlist, loitering, abandoned object, crowd anomaly, audio anomaly) with calibrated false-positive rates;
 5. Sustains a rigorous consent, revocation, audit, and bystander-protection regime compliant with the DPDP Act 2023, the Puttaswamy (2017) privacy jurisprudence, the BNSS and BSA 2023 evidentiary requirements, and forthcoming MHA / MeitY guidance on CCTV-based investigation;
 6. Integrates pluggably and optionally with CCTNS, Vahan, Sarathi, Dial 112 ERSS, and Smart-City ICCCs — without depending on any of them for core operation.
- This challenge seeks the integrated build.

3. Foundational Compliance Layer (Mandatory)

Every track and every deliverable must satisfy the following:

Principle Requirement

Citizen-pull onboarding

The system only ingests cameras that a verified citizen has explicitly registered. No discovery scans of arbitrary networks. No purchase of vendor camera lists. No “automatic” onboarding of any device.

Per-camera consent

Consent is per camera, per mode (event-only vs live-pull-enabled), per granularity operating hours, per alert type. Not bundled.

One-tap revocation,

propagated
in $\leq C0$ s

The citizen app's revocation button immediately stops all ingestion, suspends live tunnels, blocks new searches against the camera's archive, and is verifiable from the citizen's side.

Bystander
protection

At onboarding, the citizen must validate the camera's field of view. Privacy-zone masking (polygons drawn by the citizen to permanently black out areas — neighbour's window, private balcony) is mandatory and enforced both at edge and at server. Pointing a participating camera into another person's residential interior is a disqualifying configuration.

Purpose
limitation

Hard-coded use cases. The system may be used only for (a) crime investigation under a registered FIR or PCR/Daily-Diary entry, (b) missing-person search, (c) immediate-response alerts as configured, (d) traffic-incident response. Use for tax enforcement, civil disputes, employment screening, immigration enforcement, political-rally monitoring, or generic "intelligence" is structurally forbidden — not policy-forbidden, structurally forbidden (no API path).

Face
recognition
is lead-only

Every face match returns top-N candidates with calibrated probabilities. No single-match output anywhere in the UI. Every match logged with a mandatory officer attestation. No autonomous action triggered by FR.

Watchlist
governance

Watchlist entries require named approval by a Senior Superintendent of Police (or equivalent) plus a documented basis (FIR number for wanted, NCMC/Khoya-Paya reference for missing). Time-limited entries with mandatory periodic review. Categories restricted to: (i) accused with active warrant, (ii) missing persons, (iii) BOLO suspects with documented investigation linkage. Activists, journalists, protestors, political figures, and category-defined groups are categorically excluded.

Demographic
fairness

Equal Opportunity Difference and Demographic Parity Difference ≤ 10 percentage points across sex, age band, skin tone (Fitzpatrick I–VI or Monk Skin Tone), and primary region. Disparities greater than this must be disclosed and mitigated.

Calibrated
outputs

Expected Calibration Error (ECE) ≤ 0.05 for every shipped classifier. Miscalibrated models cannot ship to the live PCR console.

Tamper-
evident
chain of
custody

Every clip, every event, every search query, every live-pull session is hash-chained into an append-only audit log. Logs are externally attestable (e.g., daily Merkle-root publication).

Human-in-
the-loop

No autonomous police dispatch, alert escalation, or watchlist write from any AI output. An officer always clicks.

Bystander
erasure

Subjects of incidentally-captured footage may petition for deletion of footage of themselves under the DPDP Act; the system must support such requests with a documented workflow within mandated timelines.

Public
transparency

Aggregate, de-identified deployment statistics (camera count by area, alert volumes, search volumes, watchlist sizes, false -positive rates) must be publishable quarterly. Sunshine is the cheapest protection against mission creep.

A team that builds a brilliant detector but fails §3 will lose more points than a team that builds a modest detector and treats §3 as the architecture. Judges have been briefed accordingly.

4. Challenge Tracks

Track Focus Primary Deliverable

T1

Citizen-side: app, Bridge Agent, camera-agnostic adapter

Working onboarding flow +
ingest from ≥ 5 camera
classes

T2

Hybrid ingestion pipeline: event clips + live pull-on-demand

End-to-end clip pipeline +
WebRTC live-pull with audit
log

T3

Real-time analytics: motion, ANPR, face
detection/recognition, person/vehicle re-ID,
behavior, audio

Pluggable inference graph

with calibrated outputs

T4

PCR console: map-first investigation, search,
retrieval, trajectory reconstruction
Operator console + search
engine + multi-camera
timeline

T5 Governance: consent, revocation, watchlist, audit,
bias, bystander protection, integrations
Governance microservice +
integration adapter scaffold
Cross-track integration is explicitly rewarded. The most defensible submissions will
demonstrate end-to-end flows that traverse all five tracks.

5. Core Technical Objectives

A. Citizen-Side: App, Bridge Agent, Camera-Agnostic Adapter

A.1 Citizen mobile application (Android primary, iOS encouraged):

Onboarding flow with progressive disclosure: identity verification (Aadhaar-OTP optional, mobile-number OTP mandatory; the citizen's identity is not propagated to the PCR — only a pseudonymous Citizen-ID is), camera registration, FOV validation, privacy-zone marking, operating-hours selection, alert subscription.

Aadhaar use is opt-in only; the system must function with mobile-number-only verification. UIDAI guidelines for Aadhaar-OTP authentication must be followed where used, with auth-only flows (no demographic fetch).

Privacy-zone editor: a touch-friendly polygon tool over a still frame from the camera. Pixel-level masks applied both at the edge and at the server. Mask edits propagate within 60 s.

Away-mode toggle: lets the citizen mark times when premises are unoccupied, raising alert sensitivity for that camera. Defaults to off.

Live-pull pre-authorization toggle: per camera, per category (always-deny, ask-each-time, auto-allow for declared emergencies only).

Dashboard: list of registered cameras, current state (online/offline/paused), live-pull session history, event-clip upload history, alerts received, current participation badge / recognition (cautious gamification — see §10 constraints).

One-tap pause and one-tap revoke with confirmation. Pause is reversible; revoke triggers a U0-day deletion countdown of the camera's existing archive (configurable per jurisdictional retention policy).

Transparency feed: who accessed the citizen's feeds, when, under what authorization, and for which case reference.

A.2 Bridge Agent — the software that runs on the citizen's side and normalizes their camera into the platform's canonical interface. Reference deployment modes:

Mobile-resident (smartphone running the citizen app does ingestion for phone-cameras and locally-bonded Wi-Fi cameras when the app is active).

Router-resident (an OpenWrt package or Docker container on a home/SME router — recommended for always-on premises cameras).

Edge-box-resident (a low-cost dedicated device: Raspberry Pi 4 / 5, Orange Pi, Rockchip RK3588 boxes, NVIDIA Jetson Nano / Orin Nano for users wanting on-device analytics).

PC-resident (Windows / Linux desktop app for users with existing DVR/NVR setups).

The Bridge Agent must:

1. Discover and onboard heterogeneous cameras (see A.3).
2. Normalize to a canonical stream (H.264 or H.265 video, AAC or Opus audio, fragmented MP4 segments).
3. Run lightweight edge inference for event triggers (see §C).
4. Buffer pre-roll (5–10 s) so triggered clips include the moments before the trigger.
5. Upload event clips opportunistically over bandwidth-aware HTTPS with resume.
6. Maintain a WebRTC or SRT-tunnel-on-demand for authorized live pulls, opening only when authorized.
7. Enforce privacy zones at the pixel level before any frame leaves the citizen's device.
8. Honor consent state in real time — a revocation at the app must reach the agent within 60 s and immediately stop all uplink.
- U. Self-update securely with signed releases and a documented update channel.

A.3 Camera-agnostic adapter matrix — the Bridge Agent must support at minimum:

Class Examples Ingestion path

ONVIF

Profile S/T

Hikvision, Dahua, Axis, Bosch, CP Plus,

Prima, Uniview

ONVIF WS-Discovery +

RTSP pull

Generic

RTSP

Most IP cameras, IP doorbells Direct RTSP URL +
credentials

Vendor cloud

APIs

Tapo Care, Mi Home, Wyze, Ring, Nest,

Eufy, Reolink

Vendor-OAuth + cloud relay

DVR/NVR

(analog →

digital)

Hikvision iVMS, Dahua DMSS, CP Plus on
legacy DVRs

DVR ONVIF/RTSP backplane

or NVR SDK

Phone

cameras

Android & iOS via the citizen app

WebRTC publish

Dashcams 70mai, Garmin, Xiaomi, Pioneer — those

with Wi-Fi or companion app data export
Vendor SDK or on -trip clip
upload from companion app
USB /
webcam

Older / DIY rigs V4L2 (Linux), Media
Foundation (Windows)

For consumer-cloud cameras (Ring, Tapo, etc.), the platform must operate as a read-only OAuth client authorized by the citizen — never bypassing vendor terms. Where a vendor API does not exist, the Bridge Agent may use RTSP-over-LAN if the vendor exposes it (Tapo and most Hikvision/Dahua devices do).

A.4 KYC and identity-pseudonymization. Citizens are verified once (mobile-OTP minimum) and issued a stable but pseudonymous Citizen-ID. The PCR operator console never sees the citizen's name, phone, or address by default — only the camera's geo-location, FOV polygon, and Camera-ID. Identity linkage requires a documented, role-gated, and audit-logged “owner contact” action (e.g., the IO needs to ask the citizen for a longer time-window of footage).

B. Hybrid Ingestion: Event Clips & Live Pull-on-Demand

B.1 Event-clip pipeline (default mode).

Edge triggers a clip: 5–10 s pre-roll + event window + 10–15 s post-roll, capped at 60 s per clip (longer events generate adjacent clips with continuity tokens).

Encoding: H.265/HEVC preferred (bandwidth) with H.264 fallback, fragmented MP4, GOP-aligned at clip boundaries.

Metadata per clip: camera-ID (pseudonymous), geo-location (with citizen-set precision floor — e.g., ± 25 m to protect exact home location), timestamp (monotonic + UTC), trigger type, edge-inference outputs with confidences, hash, signature.

Upload protocol: HTTPS multipart with chunked resume; backpressure-aware (back off when uplink congested); priority-tagged (audio-anomaly and away-mode-motion clips uploaded ahead of routine clips).

Bandwidth budget: typical edge upload ≤ 200 KB/clip at 720p H.265, ≤ 800 KB/clip at 1080p. The system must function on 1 Mbps sustained uplink with bursts.

B.2 Live pull-on-demand pipeline.

An authorized PCR operator initiates a live-pull request from the console, citing case reference (FIR number, PCR-call ID, or missing-person reference). The request enters an authorization workflow:

Tier-1 events (active 112 call, active pursuit, abducted-child alert): pre-authorized for ≤ 15 -minute sessions, logged and post-reviewed.

Tier-2 events (registered FIR, named investigation): require shift-supervisor approval before tunnel opens.

Tier-3 events (intelligence-led, no FIR): require Senior SP approval and produce a flagged audit entry.

The Bridge Agent opens a WebRTC tunnel via the platform's TURN relay (citizen IP never exposed to PCR), streams at requested resolution up to camera native, and auto-terminates at session end or after a configurable max duration.

The citizen receives a real-time notification: “Police accessed your camera at 18:42 for 6 minutes under case .” The notification is non-dismissible until acknowledged and is permanently visible in the citizen's history.

For citizens who toggled “ask each time”, the tunnel does not open until the citizen approves on their phone (with a 30-second timeout for tier-1, deny-by-default).

All live frames carry a server-side watermark with session-ID for chain of custody.

B.3 Stream-protocol layer.

WebRTC (Pion / mediasoup) for live pulls — lowest latency, NAT-traversable, browser-renderable in the console.

SRT (Haivision libsrt) as fallback for resilient streaming over lossy mobile uplinks.

HLS/LL-HLS for archive playback in the console.

RTSP/RTMP only inside the citizen's LAN, never on the public network.

B.4 Edge-server protocol governance. All Bridge-Agent ↔ server traffic uses mTLS with per-agent certificates issued at onboarding, rotated quarterly. Certificate revocation propagates within 5 minutes of citizen revocation.

C. Real-Time Analytics

The analytics pipeline is hybrid edge-server: lightweight detection at the edge gates uplink; heavier analytics (ANPR OCR, face recognition, Re-ID, behavior) run on the server against uploaded clips and on-demand live frames.

C.1 Motion and presence (edge).

Algorithmic baseline: MOG2 / KNN background subtraction with per-zone activation polygons.

Learned baseline: YOLOv8-nano / YOLOvU-tiny / RT-DETR-nano for person/vehicle/animal class gating.

Outputs: bounding boxes, class, confidence, in/out of zone, line-crossings.

Targets: ≥ 12 fps on Raspberry Pi 4 / RK3588; ≥ 25 fps on Jetson Orin Nano.

Night-mode-aware: IR illumination detection and contrast normalization; reduced false-positive thresholds during the citizen's declared "away" windows.

C.2 Automatic Number-Plate Recognition (ANPR) — server-side.

Two-stage pipeline: plate-detection (YOLOv8 / RetinaNet trained on Indian plate variants — white/yellow/green/red/blue) + plate-OCR (CRNN, PaddleOCR, EasyOCR, or a transformer-based recognizer).

Indian-plate constraints:

Standard format XX-DD-XX-DDDD or XX-DD-X-DDDD (state-RTO-series-number).

Multi-script: Devanagari (Hindi), Tamil, Bengali, Kannada, etc. on commercial vehicles in some states — fall back gracefully.

High glare on reflective plates, common dust occlusion, frequent non-standard fonts on private vehicles.

Output: plate string, character-level confidence, plate-class

(commercial/private/EV/government), vehicle attributes from co-located vehicle detector (make-model-color via OSNet-Vehicle or learned classifier).

Vahan integration is optional and pluggable (\$H). Without Vahan, the system surfaces the plate string only.

KPI: ≥ U2 % character-level accuracy on a curated Indian-plate test set at ≥ 60 px plate height; ≥ 80 % at 40–60 px.

C.3 Face detection and recognition — server-side, governed.

Detection (always-on for indexed search): RetinaFace / SCRFD / YuNet — outputs bounding box and quality score. Detections produce embeddings stored as a searchable

vector index. Embeddings are anonymous — no name attached unless and until a recognition event against an authorized watchlist matches.

Recognition (only against authorized watchlists): ArcFace / AdaFace / CurricularFace embedding model. Watchlist embeddings stored in a separate, more tightly-access-controlled index.

Governance binding:

Recognition only runs against entries in §3-compliant watchlists.

A match returns top-5 candidates with calibrated probabilities — never a single name.

Confidence threshold tuned so that false-match rate ≤ 1 in 10,000 at the chosen operating point.

Mandatory officer attestation: every recognition surfaced requires the IO to click “I am visually verifying this match” before any downstream action.

Every recognition event is logged with input frame hash, candidate list, scores, attestation, and case linkage.

Bias testing: required across Fitzpatrick / Monk skin-tone scale, age bands (with children excluded from any recognition pipeline), and sex. Indian-population evaluation set required — Western-only validation is disqualifying.

Children: persons assessed as under 18 by a face-attribute classifier are excluded from recognition output entirely. Detections are still indexed but recognition matches are filtered.

C.4 Person Re-Identification (Re-ID) — server-side, investigation-bound.

Models: OSNet, FastReID, TransReID, or CLIP-ReID.

Use case: an officer selects a person crop from one clip and asks “where else does this person appear in this case’s geo-and-time window?”

Strictly bounded: Re-ID may only be invoked against the camera and time-window scope of an active investigation. Re-ID against the entire archive is disqualified by design (no API path).

Outputs ranked candidate appearances with confidence; officer reviews and confirms.

C.5 Vehicle Re-ID — server-side.

Models: parallel to person Re-ID — OSNet-Vehicle, CLIP-Vehicle.

Use case: a getaway vehicle was glimpsed on one camera; find other appearances in the locality during the window.

C.C Behavior, audio, and scene events.

Loitering: time-in-zone exceeding a per-camera threshold (citizen-configurable).

Abandoned object: dwell-time detector for unattended bags / packages (SlowFast or transformer-based action models).

Crowd anomaly: density estimation (CSRNet, MCNN) plus motion-vector divergence for sudden surges.

Fight / scuffle: action classifier (X3D / MViT / VideoMAE) with high precision threshold to limit false positives.

Audio anomaly: PANN / YAMNet / AST-based classifiers for gunshot, scream, glass break, vehicle crash. Bandwidth-friendly: edge runs the audio classifier; server confirms on uploaded clip.

All behavior outputs surface with explicit uncertainty; none should be presented as “this is a fight” without an attached probability.

C.7 Alert taxonomy. The system emits a defined and bounded set of alerts:

Alert Trigger Default audience
Away-mode
night motion
Person/vehicle in zone during
citizen-declared absent window

Citizen + PCR (citizen-opt-in)
Vehicle of
Interest
match

ANPR matches BOLO list

PCR investigation desk
Person of
Interest
match

FR top-1 match exceeds threshold
against authorized watchlist

PCR investigation desk (with
mandatory attestation gate)

Loitering Dwell-time exceeded in monitored
zone
Citizen (and PCR if zone is
opted-in public)
Audio
anomaly
Gunshot / scream / glass-break
classifier above threshold
PCR + Dial 112 (via integration
adapter)
Crowd
anomaly

Density / divergence above threshold PCR (for public-area cameras
only)
Camera
offline /
tamper

Lost connectivity > 5 min or sudden
FOV change

Citizen + PCR (operational)
Privacy-zone
violation
Edge detector flags possible mask
circumvention

Citizen (urgent)

The alert list is exhaustive for the POC. Adding new alert types requires a documented

governance step; this is by design.

D. PCR Console: Map-First Investigation

D.1 Map view (primary surface).

Vector map of the jurisdiction (MapLibre GL with on-prem tile server; OpenStreetMap base + agency overlays).

Camera pins coloured by state: online / offline / paused / event-flagged. Click to expand camera card with FOV polygon, recent events, archive timeline.

Heatmaps: incident density, alert volume, watchlist match density (last 24 h / 7 d / 30 d), coverage gaps.

Geo-fence drawing tools for ad-hoc area definitions ("show all cameras within this polygon during 18:00–20:00 on date X").

D.2 Investigation workflow.

Investigations are first-class objects. Creating an investigation requires a case reference (FIR / PCR call / missing-person ref). All subsequent searches against feeds tag back to that investigation; archives sit in an audit-isolated workspace.

Multi-camera timeline: synchronized scrubbing across selected cameras; jump-to-event; bookmark/annotate.

Search surfaces:

By attribute ("red SUV, white shirt, ~18:00–20:00, Sector 12 radius 200 m") via a structured query builder backed by the metadata index.

By plate string with fuzzy matching for OCR errors.

By face crop against the case-scoped face index (not the global archive — that is restricted).

By person crop for Re-ID.

By event type (loitering / audio / vehicle / etc.) with geo and time filters.

Cross-camera trajectory reconstruction: select a target appearance; the system proposes the next likely cameras based on geo and elapsed time and surfaces ranked candidate appearances for officer confirmation.

D.3 Operator roles (RBAC):

Role Privileges

Watcher View alerts, view public-area cameras, no archive access, no live pull

Investigating

Officer (IO)

Create investigations, search case-scoped archives, request live pull (Tier-2/3 require supervisor co-sign), attest FR matches

Shift

Supervisor

Approve Tier-2 live pulls, review investigations, sign off case closures

Senior SP Approve Tier-3 live pulls, manage watchlist entries, sign off bulk operations

Auditor Read-only access to logs, no operational privileges

System

Admin

Infra ops, no access to feed content

Two-officer rule applies to: watchlist add/remove, bulk export of case archives, citizen-

identity disclosure.

D.4 Citizen-facing PCR transparency. A read-only public dashboard (de-identified) shows aggregate platform usage so external observers can monitor for drift. Mandatory.

E. Storage, Retention, Lifecycle

E.1 Tiered storage.

Tier Content Default retention Storage class

Hot Recent event clips, live-pull recordings, alerts

14 days

NVMe / fast SSD

Warm

Older event clips

U0 days HDD / object store

Cold

Archived, only on legal hold

Case-bound Cold object store

Index Metadata, embeddings, ANPR strings, event records

1 year (configurable per jurisdiction)

Indexed DB + vector store

Audit

Append-only, hash-chained

7 years WORM-style log storage

Default retentions are floors, not ceilings; jurisdiction may shorten but should not extend without policy basis.

E.2 Legal hold. When a clip is linked to an active investigation, retention extends to case closure + statutory archival period. Hold and release are audit-logged.

E.3 Cryptographic erasure. When a citizen revokes a camera's participation, the system supports cryptographic erasure of that camera's archive (per-camera key destroyed) within the revocation deletion window (default U0 days, citizen-shortenable).

E.4 Compression and storage efficiency.

Event clips: H.265 CRF 25–28 (visually lossless for forensic use, ~5–8× smaller than H.264 baseline).

Long-tail clips down-sampled to 480p after warm-tier eviction unless flagged for case.

Embeddings stored at 512-D fp16 ($\approx$ 1 KB / face).

F. Identity-Matching Governance (Watchlists, FR, ANPR)

F.1 Watchlist data model.

```
watchlist_entry {  
  id, category (WANTED | MISSING | BoLo_SUSPECT),  
  reference (FIR_number | NCMC_ref | investigation_ref),  
  approving_officer (named), approval_timestamp,  
  reviewed_by (named), review_timestamp,  
  expiry_timestamp (mandatory, ≤ 180 days unless reviewed),  
  status (ACTIVE | EXPIRED | REMoVED),  
  removal_reason, removal_timestamp,  
  biometric_template (signed, with provenance),  
  hash_chain_anchor  
}
```

Every state transition is hash-chain-logged.

F.2 Watchlist hygiene.

Periodic mandatory review: every entry must be re-affirmed at least every 180 days.

Auto-expiry on review lapse.

Per-category caps to prevent watchlist sprawl.

A prohibited-categories list is enforced at the API layer: no entries describing political affiliation, religious affiliation, caste, occupation as journalist / activist / lawyer, attendance at protest, or membership in any lawfully constituted association.

Citizen Subject Access Request mechanism per DPDP 2023: subjects may inquire about their watchlist status (subject to lawful exception under the Act for ongoing investigations).

F.3 ANPR BOLO list. Mirrors the FR watchlist's governance but for vehicle plates.

F.4 Match handling.

An FR or ANPR match generates a lead candidate event, not an alert at the citizen-camera level (citizens never know whose presence triggered a match — that knowledge stays within the PCR).

IO receives the candidate event with top-N candidates, attests, and either confirms (logged) or dismisses (also logged, with reason). Confirmation does not constitute identification — it is the start of a verification process per investigative protocol.

F.5 Threshold and operating-point management.

FR threshold pinned at false-match-rate $\leq 1e-4$ on the population-representative evaluation set, never tuned downward without sign-off.

ANPR fuzzy-match tolerance documented and reviewable.

Drift monitoring on match-rate; sudden drops or spikes trigger an automatic review.

G. Pluggable Integrations (All Optional)

Each of the following is a separate pluggable adapter with a documented contract. The system must function in their absence; presence is configuration.

System Purpose Adapter contract

CCTNS

Look up FIR details, accused records
Read-only API;
investigations linked to
FIRs; no write-back

Vahan
Resolve plate → vehicle
make/model/owner-class (not personal
data)
Read-only; bulk-rate-
limited; cache TTL
bounded
Driving-licence data (used sparingly, hit- Read-only; rate-limited;
Sarathi and-run cases only) per-query audit

Dial 112 ERSS
Push high-severity audio / crowd alerts;
receive PCR-call IDs for investigation
seeding

Two-way; SLA-
bounded

Smart-City
ICCCs
Cross-feed ICCC public-camera context
into the same map; allow citizen cameras
to complement gaps
Read-only ingestion in;
no exposure of citizen
cameras out
Khoya-Paya /
NTMS

Missing-person matching Read-only; consent-of-
family flag preserved
National
Database of Sex
Offenders
(NDSO)

Restricted watchlist source

Tightly access-
controlled; per-officer
audit
NCRB Crime in
India dashboards

Aggregate trend overlays Read-only; aggregate-
only
Vendor camera
clouds (Tapo,
Ring, etc.)

Citizen-OAuth ingestion

Per-vendor OAuth
scope minimization

Each adapter ships with a circuit breaker — if the upstream system degrades, the citizen-facing platform must remain operational with that adapter quietly disabled.

H. Bias, Fairness, Calibration, Explainability

H.1 Mandatory evaluation set. Indian-population-representative, geographically balanced, with demographic strata. Must cover at least 6 sex × age × skin-tone strata, with minimum 200 subjects per stratum for FR. Western-only benchmarks (LFW, MegaFace, IJB) are permitted for pretraining benchmarks only; not for shipping-decision evaluation.

H.2 Required reports.

Per-classifier calibration plot + ECE.

Per-classifier demographic disparity table.

ANPR accuracy by plate-script, plate-condition, illumination.

FR ROC + DET curves by demographic stratum.

Drift dashboard with thresholds.

H.3 Explainability.

Every alert has a “why” panel: the underlying detection, the model version, the confidence, the contributing features.

Saliency / Grad-CAM overlays for FR and behavior matches.

Counterfactual hints (“would not have triggered if plate-height was < 35 px”).

I. Security, Privacy, Legal Compliance

I.1 Data residency. All PII, all biometric templates, all video archives, and all PCR-side processing are on agency or sovereign cloud infrastructure within India. No cross-border transfer of citizen-feed data.

I.2 Cryptography.

Bridge Agent ↔ ingestion: mTLS 1.3, per-agent certificates.

Live-pull tunnel: WebRTC DTLS-SRTP with E2E session keys.

Storage at rest: AES-256-XTS full-disk; per-camera content keys wrapped by a KMS / HSM.

Watchlist templates: stored encrypted with separate KMS scope; access requires named-officer credentials.

Audit logs: hash-chained, periodic Merkle-root anchored.

I.3 Identity and access.

Multi-factor auth for all PCR roles (TOTP + device-bound).

Hardware-token-bound roles for Senior SP and Auditor.

Session-time-limited; sensitive operations require step-up auth.

Citizen accounts protected by mobile-OTP + optional biometric (device-local only).

I.4 Legal mapping (mandatory deliverable). The architecture document must contain a section-by-section mapping from each technical component to:
Constitution of India, Articles 1U and 21 (with Puttaswamy (2017) 10 SCC 1 on privacy).

DPDP Act 2023 — data fiduciary obligations, consent, purpose limitation, breach notification, Significant Data Fiduciary obligations if applicable.
Bharatiya Nyaya Sanhita (BNS) 2023 — relevant offence definitions.

Bharatiya Nagarik Suraksha Sanhita (BNSS) 2023 — Sections governing search, seizure, and electronic-evidence procedure.
Bharatiya Sakshya Adhiniyam (BSA) 2023 — Section 63 electronic-records admissibility with Section-65B-equivalent certification.
IT Act 2000 (as amended) — Sections 43A, 66E, 72A; Intermediary Guidelines 2021 as applicable.
Indian Telegraph Act 1885 — to the extent any captured audio implicates intercept norms.
Aadhaar Act 201C with UIDAI Authentication Regulations — for any optional Aadhaar use.
Juvenile Justice Act 2015 — if any subject is or may be a minor.

MHA / MeitY CCTV guidelines — current advisories.

Telecom Cybersecurity Rules 2024 — for connectivity components.

I.5 Section-C5B-equivalent (now BSA s.C3) certification. Every clip exported for evidentiary use must carry the required electronic-record certificate auto-generated by the system, with the officer's named attestation.

C. Key Performance Indicators (KPIs)

KPI Target

1

Heterogeneous camera onboarding ≥ 5 distinct classes (per §A.3) onboarded end-to-end

2

Onboarding time (median, per camera) ≤ 4 minutes including FOV / privacy-zone setup

3

Revocation propagation ≤ 60 s from citizen tap to ingestion-stop, verifiable

4

Event-clip upload size ≤ 200 KB @ 720p / ≤ 800 KB @ 1080p,
H.265, 30-s clip
5 Live-pull session establishment time ≤ 8 s pU5 once authorized

6 ANPR character-level accuracy $\geq$ U2 % at plate-height $\geq$ 60 px; $\geq$ 80 % at 40–60 px

7 FR top-1 accuracy on Indian-population eval set

$\geq$ U5 % at FMR $\leq$ 1e-4 (gating)

8 FR demographic-parity difference $\leq$ 0.10 across strata (gating)

U Person Re-ID Rank-1 $\geq$ 65 % on the curated case-scope eval set

10 Vehicle Re-ID Rank-1 $\geq$ 70 % on the curated case-scope eval set

11

Audio-anomaly precision @ U5 % recall $\geq$ 0.85 on curated gunshot/scream/glass test set

12 Expected Calibration Error per shipped classifier

$\leq$ 0.05 (gating)

13

Alert end-to-end latency $\leq$ 15 s from event at camera to PCR alert surfacing

14 Map-load with 10,000 cameras $\leq$ 3 s, smooth pan/zoom

15 Audit-log tamper-detection (red-team) 100 %

16

Bystander-revocation deletion 100 % within configured window, externally verifiable

17 Two-officer-rule enforcement (red-team)

100 %

18 Watchlist prohibited-category injection (red-team)

100 % rejection

KPIs 7, 8, 12, 15, 1C, 17, 18 are gating. A submission that fails any of them cannot claim the corresponding track as complete.

7. Hackathon Deliverables

1. Working prototype demonstrating: end-to-end onboarding of $\geq$ 5 heterogeneous cameras, hybrid ingestion with both event clips and live pull, PCR console with map + search + investigation, calibrated FR/ANPR/Re-ID against a mock-watchlist, full consent-and-revocation flow.

2. Mock-deployment dataset: $\geq$ 20 cameras across $\geq$ 3 environments (residential, market, road), $\geq$ 48 hours of operation with consenting volunteer subjects, with curated event ground truth for KPI evaluation.

3. Citizen mobile app (Android, minimum), with at least the onboarding, dashboard, transparency feed, pause/revoke, and notification flows working.
4. Architecture document (≤ 30 pages).

5. Legal mapping document (≤ 20 pages) per §I.4.

6. Bias, calibration, and fairness report per §H.

7. Threat-model and red-team report per §I3.

8. Source code with documented OSS licence.

U. Live demo (20 min): onboard a new camera live in front of judges, trigger an event, watch it surface in PCR, perform a search, demonstrate a revocation.

10. Recorded demo video (≤ 10 min).

8. Judging Criteria

Dimension Weight

§3 + §F + §I — governance, consent, watchlist hygiene, bystander protection, legal mapping

30 %

Camera-agnostic onboarding (real heterogeneity, not 5 of the same vendor) 15 %

Hybrid streaming pipeline (event + live-pull, audit-clean) 10 %

Analytics quality and calibration (per-modality and integrated) 15 %

PCR console UX — does it accelerate an investigation that a paper-and-knock workflow would take days for?

10 %

Search & retrieval quality (attribute / plate / face / Re-ID) 5 %

Pluggable-integration architecture (interfaces clean, system works without them) 5 %

Code quality, documentation, reproducibility 5 %

Innovation beyond scope (stretch goals) 5 %

A submission that scores zero on the governance dimension cannot win regardless of other scores. Published in advance to set the engineering culture.

9. Suggested Tech Stack (non-binding)

Bridge Agent: Go (pion/webrtc , aler9/mediamtx), Rust (gstreamer-rs), or Python (gstreamer , aiortc). Edge inference via ONNX Runtime, TensorRT, OpenVINO, or NCNN.

Camera adapters: python-onvif-zeep , libonvif , libcurl-rtsp , vendor SDKs where required.

Streaming infra: MediaMTX or LiveKit for WebRTC; SRT-via-libsrt; HLS via Bento4 or fmp4.

Ingest service: Go or Rust microservices; Kafka or Redpanda for event bus.

Detection: YOLOv8/vU/v10, RT-DETR, MMDetection.

Tracking & Re-ID: ByteTrack, BoT-SORT, OSNet, FastReID, TransReID.

ANPR: PaddleOCR, EasyOCR, custom CRNN; YOLO-based plate detector with Indian-plate fine-tune.

FR: InsightFace (RetinaFace + ArcFace), AdaFace, SCRFD; calibration via Platt / isotonic.

Behavior / Action: MMAction2 (SlowFast, X3D, VideoMAE), audio via PANN / YAMNet / AST.

Vector index: Milvus, Qdrant, FAISS with HNSW.

Metadata DB: PostgreSQL (with PostGIS for geo) + Elasticsearch / OpenSearch for time-series event search.

Object storage: MinIO (on-prem S3), Ceph for scale.

Map/Geo: MapLibre GL + tile server (Tegola / TileServer-GL), OpenStreetMap base, OSRM for routing.

Backend: Go / Rust services; gRPC internally; REST/GraphQL externally.

PCR console: React + MapLibre + WebRTC; Electron wrapper for desktop where needed.

Citizen app: Flutter or Kotlin/Compose; WebRTC native libs.

Audit log: hash-chained append-only store (Trillian, sigstore-style transparency log).

KMS / HSM: HashiCorp Vault, AWS CloudHSM or sovereign equivalent.

Observability: OpenTelemetry, Prometheus, Grafana, Loki.

10. Stretch Goals

Coverage-gap recommender: identify “dark” street-segments based on incident density and camera positions, suggest gap areas for RWA outreach (recruitment, not coercion).

Citizen-recognition badge system with extreme care — gamification that doesn’t create a participation arms race. Recognition only, no preferential treatment, no monetary incentive. (See §11 constraints.)

Cross-jurisdiction federation: enable lawful sharing of search results across districts/states with citizen-visibility maintained.

Mobile-PCR app for IOs in the field — receive alerts, request live-pull from the scene, with stricter authorization since field requests have less oversight.

AI-assisted FOV-validation: the Bridge Agent suggests appropriate privacy zones by detecting likely private interiors / neighbour windows during onboarding.

Drone-cam temporary onboarding for citizen drones during events / disasters with explicit single-event scope and auto-revoke.

Reverse-tip-line: when an investigation needs footage from an area, the system can ask citizens in that area (without revealing the investigation) “do you have footage from at ?” with their choice to share or not. Pull, not push.

Federated-learning ANPR / behavior model improvement without raw-data sharing across jurisdictions.

Citizen-shared dash-cam triggers integrated with hit-and-run alerts when a passing vehicle matches the BOLO.

Forensic-grade clip exporter: produce BSA s.63-compliant evidence packages with hash chain, certificate, and reproducibility hash.

11. Constraints & Explicitly Out-of-Scope

The following are structurally forbidden — no API path, not a policy preference:

No unconsented camera ingestion of any kind. No scraping. No “default-on” anything.

No FR or Re-ID against the global archive. Recognition is watchlist-bound; Re-ID is investigation-bound.

No autonomous police action triggered by any AI output. Always-human-in-the-loop.

No watchlist entries describing political affiliation, religious affiliation, caste, journalistic / activist / legal-practitioner status, or protest attendance.
No cross-border transfer of citizen-feed data or biometric templates.

No use for traffic enforcement (challan generation). Different system, different governance.
No use for tax / GST enforcement, immigration enforcement, or non-criminal civil disputes.
No “predictive policing” use of stored data — the system observes and retrieves; it does not forecast individual behavior.
No monetary incentives for citizens to onboard cameras. Recognition only.

No preferential PCR response based on whether a complainant participates in the camera-sharing programme. Service is equal.
No “in exchange for” arrangements with utility services, RWA fees, or municipal services.
No use of dark-pattern UX in the consent flow. Consent options must be visually equivalent in prominence to refusal.
No always-on audio capture without separate, distinct citizen consent with default-off.
No retention beyond stated limits without case-bound legal hold.

No closed-source FR or ANPR models as black-boxes shipping into the PCR — model cards, training-data provenance, and bias audits required.

12. Timeline (indicative)

Phase Duration
Milestone
Kickoff & team
formation

Week 0 Teams registered, tracks chosen, ethics protocol
submitted
Volunteer protocol
approval

Week 1

Consent forms, mock-deployment plan approved

Design submission

Week 3 Architecture document v1, legal mapping v1, threat
model v1

Mid-point checkpoint

Week 5 Citizen-app onboarding + 3 camera classes ingesting
+ basic PCR map
Analytics + governance
freeze

Week 7 FR/ANPR/Re-ID calibrated; watchlist governance live;
bias-report draft

Red-team week

Week 8 Tamper-evidence, prohibited-watchlist, two-officer-rule tested

Field test & demo day

Week U Live demo with judges; mock investigation walkthrough

13. Threat Model & Adversarial Considerations

T1 — Citizen-side compromise. Attacker takes over a Bridge Agent to leak feeds or inject fake events. Mitigations: per-agent certs, signed update channel, attestation, anomaly detection on uplink patterns.

T2 — Police-side over-reach. An officer attempts to query outside scope (no FIR, political target). Mitigations: investigation-scoping at API layer, prohibited-category enforcement, two-officer rule, auditor role.

T3 — Bystander harm. A citizen's well-meaning camera captures a domestic-violence survivor or a stigmatized individual in compromising context. Mitigations: FOV validation, privacy zones, default detection-only (no recognition for non-watchlist faces), bystander-erasure workflow.

T4 — Stalking via the platform. A malicious actor in any role attempts to track a specific individual. Mitigations: Re-ID strictly investigation-bound, audit-log anomaly detection on search patterns, no global-archive face search.

T5 — Watchlist poisoning. Bad actor adds political target to watchlist. Mitigations: named-approver requirement, periodic review, prohibited-category API enforcement, public aggregate transparency.

TC — Model bias causing wrongful targeting. FR biased against certain demographics produces disproportionate matches. Mitigations: SH bias audit, gating KPIs, attestation requirement, top-N rather than top-1, threshold management.

T7 — Spoofed feeds. Adversary feeds synthetic or replayed footage to frame someone. Mitigations: per-agent attestation, monotonic timestamps, scene-consistency checks, video-provenance signing where available.

T8 — Live-tunnel hijack. Attacker intercepts a live-pull session. Mitigations: mTLS + DTLS-SRTP, per-session keys, citizen-side audible/visible indicator, server-side watermark.

T9 — Vendor-cloud dependency failure. A consumer-camera vendor changes API or terms. Mitigations: adapter pattern with graceful degradation, citizen notification, fallback to RTSP-LAN where available.

T10 — Mission creep. Over time, scope expands beyond the bounded use cases. Mitigations: structural — purpose-limitation enforced at the API; prohibited-categories in code; public transparency dashboard; auditor role with read access to logs; quarterly published statistics.

T11 — Court challenge. Constitutional or DPDP challenge to the deployment. Mitigations: legal mapping document, calibrated and bias-audited models, named-officer attestation, deployment-by-deployment legal review, opt-in-only architecture, demonstrated bystander protection.

T12 — Wrongful arrest based on FR. A false FR match leads to an arrest of an innocent person. Mitigations: top-N output only, mandatory officer attestation, FR is a lead, not identification; bias audit; calibration; threshold management; documented investigative protocol that requires independent corroboration before action.

14. Closing Note to Participants

This hackathon will produce useful, deployable software only if teams accept that the technology problem and the governance problem are the same problem. A citizen who shares a camera is giving the police a powerful tool and trusting that the tool will not be turned against her, her neighbour, or the next election's protestors. Honour that trust at the architecture level — not in a policy document filed somewhere — and the resulting system

will save lives, recover children, and solve real crimes. Treat it as an afterthought, and the system will be dismantled in court before it ever helps a single victim.

Build for the citizen first. The police are well served by that order.